

Distributed Robotics practical: Assessment criteria

Last updated on 15 June 2026

1 Overview

Work is to be conducted in teams. Teams are composed of 2-3 students (in exceptional cases, 4 students may be allowed). They self-allocate and then register with the teaching staff.

Topics can be proposed and adjusted in discussion with the teaching staff. As a minimum, the topics should be relevant to the theme of *distributed robotics*. They should report findings (e.g. theoretical, computational and/or real) from studies involving multiple robotic agents.

2 Report

Each team must write an **academic report** using the standard **IEEE conference template** (<http://ras.papercept.net/conferences/support/support.php>), preferably for Latex.

The report must be submitted as a **PDF file** no later than **Tuesday, 1 September 2026 at 23:59 pm**. The report can be up to **5 pages** long (excluding references and appendix) and formatted using the provided standard IEEE conference template.

Your report should be structured as follows (see Table 1, for additional information):

- **Title:** Sufficiently informative (e.g. not just “Lab report” but it is suggested to include “ (Distributed Robotics practical)” at the end)
- **Author:** Your names
- **Affiliation:** “[Your surname] is/are with the Department of (TODO), Technische Universität Darmstadt, Germany” (placed on page 1, bottom of the left column, according to IEEE conference template).
- **Abstract:** Describe in up to 100 words the content of your report (do not include irrelevant details like “Distributed Robotics Practical”, your degree program, etc).
- **Introduction** (section): Provide context and motivation for the concrete work being performed. The section will demonstrate an awareness of related work (literature). Include one concise paragraph outlining the contents of the report. E.g. “This report is organized as follows. Section 2 presents ...”.
- **Problem formulation** (section): formulate the problem that is the subject of your studies. Provide sufficient details e.g. about the environment/world, the robots (e.g. are they embodied, how do they move, what do they sense etc., the objective)
- **Design** (section): describe your proposed solutions (e.g. control strategies) at a suitably high level. Avoid ambiguity (e.g. “the robot solves

the task as efficient as possible”). Consider suitable visualizations e.g. finite-state machine diagrams or pseudocode. No source code!

- **Implementation** (section): describe how your proposed solutions are implemented. This section is meant to bridge the gap between the design section and the source code in the appendix. Ideally refer to key elements in the appendix e.g. “... (see Lines 35—40 in Appendix A)”.
- **Results** (section): describe how your proposed solutions are to be evaluated, and the findings you obtained. These can be theoretical, obtained via computer simulations, obtained via experiments with real robots, or a combination of this.
- **Discussions** (section): Discuss the findings. Include an appreciation of possible ecological and social effects.
- Do not include any conclusions section for this lab report.
- References (can be included) – do not count towards 5 page limit
- Appendix (section): Your source code properly formatted as text (not as images), with syntax highlighting and line numbers. For the appendix, use single-column format. You can provide additional appendices for supplementary materials e.g. further figures, pseudocode, tables, a proof. Note however that the appendices may not be read in detail, and the marks are primarily based on the main report, which should be sufficiently self-contained.

This is an academic report, avoid using, for instance, “I”, “my”, “us”, and “we”. An example report in IEEE format (though longer than 2 pages and including sections that you do not need) can be found here: <http://dx.doi.org/10.1109/IROS.2014.6943114>

The detailed marking criteria are listed in Table 1.

Table 1 (Marking criteria for report). Listed are the maximum points that can be achieved.

Points (out of 100)	Criteria
20	General writing quality & formatting (10) Presentation quality of report (excluding appendix). Organization (title, author name, affiliation, abstract, structure (sections and paragraphs), figures or tables where relevant); writing quality and clarity (clear and concise language, grammar, no typos, no ambiguous/informal expressions, no inconsistencies etc); adhering to standard conventions (figures/tables with captions and all cited in text [e.g. "Figure 1 shows..."], abbreviations defined when first used); use of external images acknowledged in captions (e.g. "Image reprinted from [x]"). Abstract (5) Motivating sentence, providing some context. Stating the problem that is addressed. Stating what (method) is proposed as the solution. Stating what results were obtained. And stating what is concluded from this. Appendix - source code (5) Presentation quality of appendix (e.g. appropriate commenting, as well as syntax highlighting, line numbers, consistent indentation).
20	Quality of introduction and problem formulation sections You can get up to 5 points for motivating your study. Don't just motivate something generic e.g. "robotics" or "AI" but rather the concrete problem you address. Ideally point out some technical challenges too (i.e. why is it actually a problem). You can get up to 5 points for demonstrating an awareness of related work. Refer to the works in the literature that closely resemble your own work, and point out the differences. You can get up to 10 points for formulating your problem. This can be verbally (yet not informal) but could also be supported by some mathematical objects (e.g. graphs, equations).
20	Quality of design (20) Explain the design of the proposed solutions you have chosen - <i>how</i> do they enable the robots to solve the stated problem? Choose an appropriate level of abstraction e.g. a correct finite state machine diagram, pseudocode, equations). Presenting source code or specific parameter values would be too low level (this would be <i>implementation</i>). Make sure the text body sufficiently explains any elements described in the figures (e.g. not just "Figure 1 explains the strategy.").
10	Quality of the <i>implementation</i>, including source code. You can get up to 5 points for describing <i>how</i> the designs (e.g. strategies) have been implemented. Avoid repeating what the designs are - just explain how their different elements are implemented. Do not put source code in the

	implementation section, rather point the reader to the appropriate lines of source code in the appendix. The implementation section should help the reader understand how the source code (in the appendix) relates to the design/strategy (in the previous section). You can get up to 5 points for source code that is a correct implementation of the design/strategy, well structured, and efficiently implemented. Think about whether the implementation is correct and efficient. Think about whether all lines of source code are needed. Think about using built-in commands.
30	Quality of results and discussions (including critical analysis) You can get up to 5 points for describing what situations were tested (your experimental setups for Tasks 1 and 2), and the results from your experiments in the lab - what worked, and what did not. Try to do so using unbiased, formal language, where possible based on facts (don't write "The robot did a great job."). You can get up to 20 points for providing a critical analysis of the findings obtained, and for discussing possible future improvements. These can be theoretical, computational or by experiment. Where possible ground your discussions, for example, in observations made with the robots. You can get up to 5 points for showing an appreciation of possibly ecological and social effects in potential future applications of your proposed technologies.

Penalties (Percentages of the report mark)

- 5% per day (late submissions)
- 10% penalty if source code not provided as text (but image)
- 10% penalty if report excluding references and appendix exceeds 5 pages
- 10% penalty if not using the IEEE template
- Further penalties (up to 100%) may be applied for students who regularly fail to attend scheduled teaching sessions (excluding absences that were authorized in advance).

Final marks will be derived as follows (after penalties have been applied):

90% - 1.0
 85% - 1.3
 80% - 1.7
 75% - 2.0
 70% - 2.3
 65% - 2.7
 60% - 3.0
 55% - 3.3
 50% - 3.7
 45% - 4.0
 <45% - 5.0 (ungenügend)

3 Bonus

Teams can be awarded a “bonus” of up to 0.4 (e.g. from 2.7 to 2.3, or from 2.0 to 1.7). To apply for a bonus award, the team needs to submit a video recording by **Tuesday, 1 September 2026 at 23:59 pm. A bonus will only be awarded if the groups achieved a pass mark (4.0 or above).**

An example video can be found here: <https://www.youtube.com/watch?v=9afhl1pgel8>

The teaching staff takes the following criteria into account to decide whether the video recording warrants the award of a bonus:

- All contents must be the students own work.
- The video recording shall be of good overall presentation quality.
- The video recording shall be at least 2 mins and at most 4 mins long.
- The video should have a title “slide” matching the title of the report as well as indicate “Supplementary video” (or similar), plus provide the author names and affiliations.
- The content of the video recording shall suitably complement the content of the report; excessive amounts of text should be avoided. Voice-over is encouraged (but not compulsory).
- While computer simulation clips can be included, there should be at least one clip showing real robots (e.g. multi-robot experiments, or attempts/discussions to port the findings from simulation to the real platform).

4 How to Deal with Conflicts with Teams

Conflicts naturally arise when working in groups, and often these are due to misunderstandings and can quickly be resolved. There will be times where a team member contributes more than another, especially if falling behind in their studies, or getting ill. It is suggested that all team members remain supportive of each other and be positive in their communications.

Where a student has concerns about the contribution made by other members of the team, the team should meet to resolve this in a calm and professional manner. It is recommended to do so via a face-to-face meeting.

Where a team does not manage to resolve a significant conflict on its own (e.g. a member not being reachable for a significant period), it is recommended to get in touch with the teaching staff early on, who will help to resolve the conflict.

Where the module leader determined that there have been significant concerns regarding the engagement of a particular team member, a penalty of up to 100% could be applied to the report mark of that individual (but not of the others). If this is the case, the module leader will consider discussions they had with any team member (where applicable) and general patterns of absence (without authorization).